MovementLogger — Sensor & Module Catalog

12 Teile · 2026-05-25 18:30 UTC · BOM aus movement_logger_firmware · via crawl2pump

12 neu (7 Tage) 0 aktualisiert

Build & Pairing — Standalone GPS Logger

Architektur: Die STEVAL-MKBOXPRO loggt ihre Sensoren weiterhin auf ihre eigene microSD; ein zweites wasserdichtes Kästchen auf dem Deck enthält einen eigenständigen GPS-Logger (OpenLog Artemis + Qwiic-GNSS-Modul + LiPo). Beide laufen unabhängig; in der Nachbearbeitung werden die Daten per Zeitstempel zusammengeführt. Keine UART-Verkabelung, kein JP2/JP4-Löten, keine Firmware-Anpassung an der STEVAL.

Genauigkeits-Stufen — wähle den GPS-Receiver

Tier	Receiver	Bänder	Genauigkeit (open sky)	Strom	Preis
m-Level	1.1 **SparkFun MAX-M10S Qwiic**	L1-only · 4 GNSS	~1.5–2 m	~12 mA	~40 CHF
m-Level (Patch)	1.2 **SparkFun NEO-M9N Qwiic**	L1-only · 4 GNSS	~1.5–2 m (onboard Patch)	~25 mA	~70 CHF
cm-Level (RTK)	2.1 **SparkFun ZED-F9P Qwiic**	**L1+L2 Multi-Band** · 4 GNSS	**~1–2 cm** mit NTRIP-Korrekturen via WiFi (gratis in CH via swisstopo SWIPOS)	~70 mA	~250 CHF

Antennen — was zu welchem Receiver passt

Wichtig: Antenne muss zum Receiver passen — sowohl beim **Stecker** (u.FL am MAX-M10S/NEO-M9N · SMA am ZED-F9P) als auch bei den **Frequenzbändern**. Ein L1-only Receiver braucht nur L1; ein ZED-F9P braucht **L1 + L2** (nicht L5!). Falscher Combo = stille RTK-Verlust.

Antenne	Stecker	Bänder	+ 1.1 MAX-M10S	+ 1.2 NEO-M9N	+ 2.1 ZED-F9P	Grösse / Mount
1.3 **Molex GPS-15246 Flex** (passiv, ~+2 dBi)	**u.FL** direkt	L1+L5	□ ja	□ ja	□ **NEIN** (kein L2, falscher Stecker)	40 × 15 × 0.1 mm · adhesiv unter Klar-Deckel
2.2 **u-blox ANN-MB-00** (aktiv, +28 dB LNA)	**SMA**	**L1+L2+L5** Multi-Band	□ ja (mit 3.1 Pigtail)	□ ja (mit 3.1 Pigtail)	□ **direkt** — die einzige passende RTK-Antenne	60 × 60 × 16 mm Puck + 5 m Koax · magnetisch

3.1 **SMA → u.FL Pigtail (CAB-09145):** Adapter zwischen ANN-MB-00 (SMA) und MAX-M10S / NEO-M9N (u.FL). Nur nötig, wenn du die ANN-MB-00 an einem L1-only Receiver betreiben willst. Für den ZED-F9P NICHT nötig (Board hat SMA direkt). Für die Molex Flex auch NICHT nötig (u.FL direkt am Board).

Empfohlene Kombinationen

Use-Case	Receiver	Antenne	Case-Stil
Kompakt, alles im Case	1.1 MAX-M10S	1.3 Molex Flex u.FL (unter Klar-Deckel)	Geschlossen, kein Gland, ~2 m Genauigkeit
Einfachste Lösung	1.2 NEO-M9N (Patch onboard)	keine extra — Stock-Patch unter Klar-Deckel	Geschlossen, kein Gland, ~1.5–2 m Genauigkeit
Beste L1-Genauigkeit	1.1 MAX-M10S	2.2 ANN-MB-00 + 3.1 u.FL→SMA Pigtail	Case mit SMA-Gland für Puck-Mount aussen, ~1 m
RTK cm-Genauigkeit	2.1 ZED-F9P	2.2 **ANN-MB-00** (Pflicht — Multi-Band L1+L2)	Case mit SMA-Gland, NTRIP-WiFi-Verbindung nötig (z.B. STEVAL-Hotspot oder eigenes WiFi)

Farbkodierung & Nummerierung im Katalog

Jede Karte hat eine **Nummer** (z.B. 1.1, 2.2, 0.3) und eine **Farbe** — gleiche Farbe / gleiche erste Ziffer = passen zusammen. Wähle innerhalb jeder Gruppe einen Receiver (X.1 / X.2) und die dazu passende Antenne (X.3 für L1, X.2 für RTK):

1.1 1.2 1.3 **Blau** — L1-only u.FL Gruppe. **1.1** MAX-M10S oder **1.2** NEO-M9N + **1.3** Molex Flex Antenne (direkt am u.FL).

2.1 2.2 **Orange** — RTK Multi-Band SMA Gruppe. **2.1** ZED-F9P + **2.2** ANN-MB-00. Diese zwei gehören zwingend zusammen für cm-Genauigkeit.

3.1 **Violett** — u.FL→SMA Pigtail (Brücke). Nur nötig, wenn du die ANN-MB-00 (2.2) am MAX-M10S (1.1) oder NEO-M9N (1.2) betreiben willst — verbindet SMA-Antenne mit u.FL-Receiver.

0.1 0.2 0.3 0.4 0.5 0.6 **Grau / Weiss** — bauunabhängige Teile (0.1 Host, 0.2 Akku, 0.3 **JST-PH 2-pin Strom-Kabel** für LiPo-Verlängerung, 0.4 Case, 0.5 optionaler IMU, 0.6 **Qwiic-Daten-Kabel** für die Sensor-Daisy-Chain).

Wichtig — 0.3 ≠ 0.6: 0.3 ist ein 2-pin **Strom**-Kabel (2.0 mm Pitch, dicker) zum Verlängern des LiPo-Leads. 0.6 ist ein 4-pin **Daten**-Kabel (1.0 mm Pitch, schmaler) zwischen den Qwiic-Sensoren. Verschiedene Stecker, verschiedene Anwendung.

Einkaufs-Beispiele:
Kompakt-Build = 0.1 + 1.1 + 1.3 + 0.2 + 0.3 + 0.4
Beste L1 = 0.1 + 1.1 + 2.2 + 3.1 + 0.2 + 0.3 + 0.4
RTK = 0.1 + 2.1 + 2.2 + 0.2 + 0.3 + 0.4
Mit Cross-Validation IMU = jeder Build oben + 0.5 Qwiic-IMU + 0.6 extra Qwiic-Kabel (ST-Schwester-Chip der STEVAL-IMU, für Sync-Validation oder als Backup).
Qwiic-Kabel-Bedarf: 1× 0.6 pro Verbindung in der Daisy-Chain. Artemis + 1 Sensor = 1 Kabel. Artemis + GPS + IMU = 2 Kabel. Plane danach.

Aufbau — Plug-and-Play, kein Löten

1. 0.1 **OpenLog Artemis** ist der Host: USB-C Strom + Konfig, microSD-Slot, JST-PH 2-pin LiPo-Eingang, BLE über Apollo3 Blue. **Wichtig:** diese aktuelle DEV-19426 Revision hat **keinen IMU und keinen Barometer onboard** (SparkFun hat beide aus Lieferketten-Gründen entfernt). Für Bewegungsdaten musst du einen separaten Qwiic-IMU dazustecken (z.B. ICM-20948 PRT-15335, BNO086 PRT-22857). Für unsere Architektur OK — die STEVAL trägt eh schon alle Sensoren, der Artemis ist hier nur GPS-Logger.
2. **GNSS-Modul** deiner Wahl (1.1 1.2 2.1) via 0.6 **Qwiic-Kabel** (JST-SH 4-pin, NICHT verwechseln mit dem 2-pin LiPo-Kabel 0.3) am Artemis-Qwiic-Port — automatisch erkannt von der OpenLog_Artemis Firmware. Update-Rate, Felder (Lat/Lon/Höhe/Speed/Satellites) im seriellen Menü konfigurierbar.
3. 0.2 **LiPo 1 Ah** (PRT-13813) in die JST-PH 2-pin Buchse am Artemis. Polarität ist out-of-the-box korrekt (rot = +). Laden via Artemis-USB-C — onboard MCP73831 Ladechip (~500 mA, ~2 h voll). Kein extra Ladegerät nötig. Laufzeit ~12–20 h bei 1 Hz Logging.
4. 0.4 **SERPAC RBF33P06C10C** Case (82 × 80 × 35 mm, IP67, Polycarbonat-Klar-Deckel): Artemis + GNSS-Breakout + LiPo passen comfortably nebeneinander. Klar-Deckel ist Pflicht für die 1.3 Molex-Flex-Antenne (Signal geht durch PC). Mit Klett oder 0.3 JST-Jumper sichern — ein flatternder LiPo unter Wellenkräften reisst Lötstellen / Stecker aus.
5. 0.5 **Optional: Qwiic IMU** (SparkFun ISM330DHCX SEN-19764) — braucht ein **zweites** 0.6 Qwiic-Kabel (Daisy-Chain: Artemis → 0.6 → GNSS → 0.6 → IMU). ST-Schwester-Chip der STEVAL-IMU — gleiches Rauschverhalten, gleiches Register-Layout. Sinnvoll wenn du STEVAL und OpenLog-Daten cross-validieren oder als Backup loggen willst. Für reines GPS-Tracking nicht nötig.
6. microSD reinstecken (8–32 GB Class 10, separat zu kaufen — nicht in der BOM). Knopf drücken, loggt CSV mit Timestamp + allen aktivierten Feldern.
7. Nach der Session: USB-C ans Mac/PC, SD-Karte als Block-Device gemountet, CSV importieren. STEVAL-CSV + GPS-CSV nach Zeitstempel mergen.

Dev Boards 3

SparkFun OpenLog Artemis (Apollo3 datalogger host, microSD, USB-C, Qwiic) OSS firmware USB-C pluggable

☑ USB-C ☐ WiFi ☑ Bluetooth ☐ GPS ☐ Motion sensors ☑ SD-card L×B×H 5.3 × 4.1 × 1.5 cm · 13 g

MCU: Apollo3 Blue · Cortex-M4F @48 MHz · 1 MB flash / 384 KB SRAM · BLE 5

Datasheet: https://www.sparkfun.com/products/19426

OSS firmware: https://github.com/sparkfun/OpenLog_Artemis

0.1 neu **[SparkFun OpenLog Artemis (Apollo3 datalogger host, microSD, USB-C, Qwiic) · DEV-19426 @ DigiKey](https://www.digikey.com/en/products/detail/sparkfun-electronics/19426/16570916)** **CHF 38.13**

SparkFun @ DigiKey · in stock

Standalone Qwiic datalogger host. Apollo3 Blue Cortex-M4F @48 MHz + microSD slot + USB-C charge + BLE 5 + 4× ADC inputs + RTC with backup battery. **No IMU or barometer onboard** — the current DEV-19426 rev shipped without the ICM-20948 / MS5637 that earlier revs carried (SparkFun supply- chain change; their product page confirms). For motion data, plug a separate Qwiic sensor in (ICM-20948 / BNO086 / ISM330DHCX) — Artemis firmware auto-detects 30+ supported Qwiic sensors. Runs SparkFun's open OpenLog_Artemis firmware out of the box: power on, it logs every Qwiic device it detects (any GPS in this BOM, plus any Qwiic IMU you add) to a CSV/UBX on the SD card. Configure sample-rate, file naming, sleep behaviour via a USB serial menu. The natural pair for a small GPS build: plug in a SparkFun ZED-F9P or MAX-M10S Qwiic breakout, hook up a LiPo, drop it in the SERPAC RBF33 IP67 case — a…

https://www.digikey.com/en/products/detail/sparkfun-electronics/19426/16570916

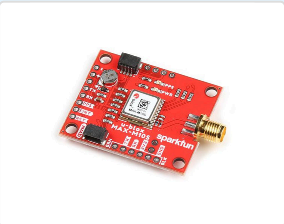
GPS / GNSS 15

SparkFun GPS Breakout - MAX-M10S (Qwiic) `OSS firmware` `Qwiic / I²C`

☐ USB-C · ☐ WiFi · ☐ Bluetooth · ☑ GPS · ☐ Motion sensors · ☐ SD-card · L×B×H 2.54 × 2.54 × 0.6 cm · 4 g

Datasheet: https://cdn.sparkfun.com/assets/7/5/9/a/a/MAX-M10S_DataSheet_UBX-20035208.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware



1.1 neu **SparkFun GPS Breakout - MAX-M10S (Qwiic) · GPS-18037 @ DigiKey** **CHF 35.80**

SparkFun @ DigiKey · in stock

Recommended MAX-M10S carrier. Open-hardware breakout. · Qwiic / I²C · firmware: open-source

https://www.digikey.com/en/products/detail/sparkfun-electronics/18037/16719314

SparkFun NEO-M9N Chip Antenna Breakout (Qwiic, concurrent GNSS L1) `OSS firmware` `Qwiic / I²C`

☐ USB-C · ☐ WiFi · ☐ Bluetooth · ☑ GPS · ☐ Motion sensors · ☐ SD-card · L×B×H 2.6 × 1.7 × 0.6 cm · 5 g

Datasheet: https://content.u-blox.com/sites/default/files/NEO-M9N_DataSheet_UBX-19014285.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware

1.2 neu **SparkFun NEO-M9N Chip Antenna Breakout (Qwiic, concurrent GNSS L1) · GPS-15733 @ DigiKey** **CHF 59.59**

SparkFun @ DigiKey · in stock

Concurrent four-GNSS receiver (GPS + Galileo + GLONASS + BeiDou, all L1) on a tiny 26 × 17 mm Qwiic breakout with onboard chip antenna. Position accuracy ~1.5 m standalone, ~0.6 m with SBAS corrections — the middle tier between the cheaper single-system MAX-M10S (slightly faster TTFF but ~2 m) and the multi-band ZED-F9P (RTK, cm-level). Update rate up to 25 Hz, ~30 mA. Plug into the OpenLog Artemis via Qwiic for a sub-meter-class standalone GPS logger without RTK complexity / subscription / NTRIP. · Qwiic / I²C · firmware: open-source

https://www.digikey.com/en/products/detail/sparkfun-electronics/15733/11481329

Molex Flexible GNSS Antenna u.FL (passive L1, MAX-M10S/NEO-M9N only) `OSS firmware` `SMA / U.FL`

☐ USB-C · ☐ WiFi · ☐ Bluetooth · ☐ GPS · ☐ Motion sensors · ☐ SD-card · L×B×H 4 × 1.54 × 0.01 cm · 2 g

Datasheet: https://www.sparkfun.com/products/15246

1.3 neu **Molex Flexible GNSS Antenna u.FL (passive L1, MAX-M10S/NEO-M9N only) · SparkFun 15246 @ sparkfun.com** **USD 6.50**

SparkFun

Passive flex GNSS-Patch (Molex 146236-0150 via SparkFun GPS-15246): 40 × 15 × 0.1 mm, adhesive backing, 50 mm u.FL pigtail — plugs straight onto the MAX-M10S / NEO-M9N Qwiic breakout's u.FL without the SMA pigtail (CAB-09145). Element covers L1 + L5/E5a/B2a — fine for L1-only receivers. **NOT compatible with the ZED-F9P** (needs L2, not L5; also the F9P breakout is SMA, not u.FL — a u.FL→SMA adapter would let it physically connect but the missing L2 band would silently kill RTK). Passive (~+2 dBi gain) — better than the breakout's onboard chip antenna, worse than a +28 dB LNA active puck (PRT-14986). The smallest no-protrusion antenna option in the BOM: stick it flat to the inside of the RBF33's clear PC lid; the signal passes through the polycarbonate. Best for case-size-constrained MAX-M10S / NEO-M9N builds where you'd rather accept a few dB of antenna loss than route an external puck…

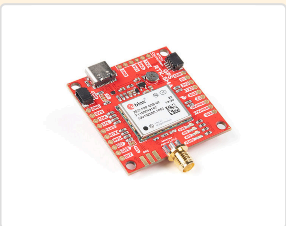
https://www.sparkfun.com/products/15246

SparkFun GPS-RTK-SMA Breakout - ZED-F9P (Qwiic, multi-band RTK) OSS firmware Qwiic / I²C

☐ USB-C ☐ WiFi ☐ Bluetooth ☑ GPS ☐ Motion sensors ☐ SD-card L×B×H 4.6 × 4.3 × 0.6 cm · 10 g

Datasheet: https://content.u-blox.com/sites/default/files/ZED-F9P-04B_DataSheet_UBX-21044850.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware



2.1 neu **SparkFun GPS-RTK-SMA Breakout - ZED-F9P (Qwiic, multi-band RTK) · GPS-16481 @ DigiKey** **CHF 214.23**

SparkFun @ DigiKey · in stock

DIY building block for a custom RTK tracker. u-blox ZED-F9P multi-band (L1+L2 GPS / Galileo / GLONASS / BeiDou) — same chip as the RTK Facet above, but as a bare Qwiic breakout for board-mount builds. Pair with any ESP32 dev board in this BOM (SparkFun Thing Plus-C, Seeed XIAO ESP32-S3) for WiFi + an NTRIP client; SparkFun ships the Arduino library `SparkFun u-blox GNSS Arduino Library` that wraps UBX protocol over I²C/Qwiic. SMA antenna jack onboard — wire to the u-blox ANN-MB-00 multi-band antenna below (L1+L2 active patch, 5 m coax, magnetic mount) for production performance. Without RTK corrections this is still a ~30 cm receiver under good sky; with RTK it's 1–2 cm. · Qwiic / I²C · firmware: open-source

https://www.digikey.com/en/products/detail/sparkfun-electronics/16481/12178437

u-blox ANN-MB-00 — multi-band active GNSS antenna (L1+L2+L5, SMA, 5 m) OSS firmware SMA / U.FL

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☐ Motion sensors ☐ SD-card L×B×H 6 × 6 × 1.6 cm · 120 g

Datasheet: https://content.u-blox.com/sites/default/files/ANN-MB_DataSheet_UBX-18049862.pdf

2.2 neu **u-blox ANN-MB-00 — multi-band active GNSS antenna (L1+L2+L5, SMA, 5 m) · ANN-MB-00 @ DigiKey** **CHF 48.07**

u-blox @ DigiKey · in stock

Multi-band L1/L2/L5 active patch antenna for the ZED-F9P. Magnetic base, 5 m RG-174 coax, SMA male plug — the antenna u-blox themselves spec for multi-band testing. Without a multi-band antenna the ZED-F9P falls back to L1-only and you lose the RTK precision advantage; pair these two. Mount the puck on a non-metal ground-plane (the foilboard deck or case lid) with clear sky view. · SMA / U.FL · firmware: open-source

https://www.digikey.com/en/products/detail/u-blox/ANN-MB-00/9817928

Interface Cable SMA → U.FL, 100 mm (pigtail) OSS firmware SMA / U.FL

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☐ Motion sensors ☐ SD-card L×B×H 10 × 0.3 × 0.3 cm · 3 g

Datasheet: https://www.sparkfun.com/products/9145

3.1 **neu** **Interface Cable SMA → U.FL, 100 mm (pigtail) · SparkFun 9145 @ sparkfun.com** — **USD 5.75**

SparkFun

100 mm pigtail: U.FL female (plugs into the SparkFun MAX-M10S breakout) ↔ SMA female bulkhead (mates with the antenna above's SMA male). Buy with the magnetic antenna or skip both if the onboard chip antenna is enough. · SMA / U.FL · firmware: open-source

https://www.sparkfun.com/products/9145

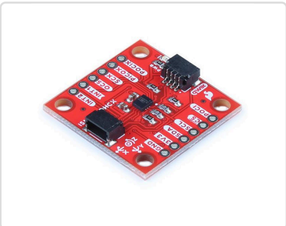
IMU (Accel + Gyro) 3

SparkFun ISM330DHCX 6-DoF IMU (Qwiic, ST industrial-grade) `OSS firmware` `Qwiic / $I^2C$`

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☑ Motion sensors ☐ SD-card L×B×H 2.54 × 2.54 × 0.6 cm · 4 g

Datasheet: https://www.st.com/resource/en/datasheet/ism330dhcx.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware



0.5 **neu** **SparkFun ISM330DHCX 6-DoF IMU (Qwiic, ST industrial-grade) · SEN-19764 @ DigiKey** — **CHF 19.86**

SparkFun @ DigiKey · in stock

ST ISM330DHCX 6-DoF IMU on a 1×1″ Qwiic breakout. Industrial-grade (-40 to +105 °C, ST 10-year part availability), accel ±2/4/8/16 g + gyro ±125/250/500/1000/2000/4000 dps, embedded Machine Learning Core + Finite State Machine for on-chip activity classification. Accel-Rauschen ~60 µg/√Hz, gyro ~3.8 mdps/√Hz — practically the same as the STEVAL-MKBOXPRO's LSM6DSV16X (ST family sibling) and ~4× cleaner than the consumer-grade ICM-20948 the OpenLog Artemis used to ship with. No magnetometer onboard — add a separate Qwiic magnetometer (e.g. MMC5983MA PRT-19921) only if you need heading; for pumpfoil pump-stroke detection it's not necessary. Plugs into the Artemis Qwiic daisy-chain alongside the GPS receiver; OpenLog_Artemis firmware auto-detects and logs both. · Qwiic / I^2C · firmware: open-source

https://www.digikey.com/en/products/detail/sparkfun-electronics/19764/16653193

Distance / ToF 3

SparkFun Qwiic Cable 100 mm (solder-free I^2C link) `OSS firmware` `Qwiic / $I^2C$`

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☐ Motion sensors ☐ SD-card L×B×H 10 × 0.5 × 0.3 cm · 2 g

0.6 **neu** **SparkFun Qwiic Cable 100 mm (solder-free I^2C link) · SparkFun 14427 @ sparkfun.com** — **USD 1.50**

SparkFun

4-pin JST-SH Qwiic / STEMMA-QT cable (100 mm). The no-solder link from the VL53L1X Qwiic breakout to the LilyGO T-Beam S3 Supreme's QWIIC socket (or the STEVAL I^2C bus). Order one alongside the ToF — the breakout does not ship with a cable. · Qwiic / I^2C · firmware: open-source

https://www.sparkfun.com/products/14427

Battery / Power 5

SparkFun Lithium Ion Battery 1Ah, JST-PH 2-pin (OpenLog Artemis power) `OSS firmware` `battery cell`

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☐ Motion sensors ☐ SD-card L×B×H 6 × 3.6 × 0.7 cm · 22 g

Datasheet: https://www.sparkfun.com/products/13813

0.2 neu **SparkFun Lithium Ion Battery 1Ah, JST-PH 2-pin (OpenLog Artemis power) · SparkFun 13813 @ sparkfun.com** **USD 16.95**

SparkFun

1 Ah single-cell LiPo (3.7 V nominal) with JST-PH 2.0 mm 2-pin connector — plugs straight into the OpenLog Artemis (DEV-19426) battery jack. The Artemis's onboard MCP73831 charges it over USB-C (~500 mA) and runs the board untethered when unplugged. Runtime at 1 Hz GPS logging ≈ 12–20 h. Polarity is correct out of the box for SparkFun loads — verify with a multimeter before connecting to anything else. Mount with foam or velcro inside the case; an unrestrained LiPo under wave load can tear the JST leads off the cell. Class 9 lithium shipping rules apply: most distributors ship surface only, no air freight to private addresses. · battery cell · firmware: open-source

https://www.sparkfun.com/products/13813

SparkFun JST Jumper 2 Wire Assembly (JST-PH 2-pin extension)

OSS firmware · battery cell

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☐ Motion sensors ☐ SD-card L×B×H 15 × 0.5 × 0.3 cm · 2 g

Datasheet: https://www.sparkfun.com/products/8670

0.3 neu **SparkFun JST Jumper 2 Wire Assembly (JST-PH 2-pin extension) · PRT-08670 @ Mouser** **CHF 0.87**

SparkFun @ Mouser · in stock

JST-PH 2.0 mm 2-pin connector (female) → 2× bare stranded wires, ~150 mm. Use it to extend the LiPo lead inside the case (battery on one side, logger on the other), or to splice in a hard power-switch for a clean on/off without unplugging the cell. Polarity matches SparkFun LiPos — red on the same pin as the cell's red lead. Solder + heat-shrink to extend; for a JST-to-JST extension, twist two of these back to back. NOT a load-bearing power link on its own — too thin for sustained > 1 A draw, fine for the Artemis's ~50–80 mA logging current. · battery cell · firmware: open-source

https://www.mouser.ch/ProductDetail/SparkFun/PRT-08670?qs=WyAARYrbSnYwIBYMDo3WjA%3D%3D

Enclosure 1

SERPAC RBF33P06C10C — IP67 PC enclosure, clear lid (82 × 80 × 35 mm)

OSS firmware · enclosure

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☐ Motion sensors ☐ SD-card L×B×H 8.2 × 8 × 3.5 cm · 65 g

0.4 neu **SERPAC RBF33P06C10C — IP67 PC enclosure, clear lid (82 × 80 × 35 mm) · RBF33P06C10C @ SERPAC (vendor page)** **USD 14.27**

SERPAC (manufacturer store) · in stock

Polycarbonate IP67 project box, external 82 × 80 × 35 mm (3.23 × 3.15 × 1.38 in), clear PC lid sealed with perimeter O-ring + stainless-steel screws into metal inserts. Smallest RBF footprint that comfortably fits the OpenLog Artemis (53 × 41 mm) + a ZED-F9P Qwiic breakout (46 × 43 mm) stacked, plus a small LiPo. Same RF-transparent / Hall-magnet / Qi-charge tricks as the RBF53 / RBF63 line. For the GPS-only standalone logger build — STEVAL stays in its own RBF53 case nearby, GPS box sits separately on the deck with clear sky view. · enclosure · firmware: open-source · manufacturer page — price/stock also via Mouser/DigiKey/Farnell

https://www.serpac.com/rbf33-electronic-enclosure.html
